```
In[72]:= Clear[cand, votes, preferance, first, isValid, errorMsg];

cand = Input["Type candidate names inside
    curly braces, separated by commas.\nExample: {p, r, s, t}"];
votes = Input[
   "Type the top row of vote counts inside curly braces.\nExample: {14, 10, 8, 4, 1}"];
preferance = Table[Input["Type the candidates for Rank " <> ToString[i] <>
    " inside curly braces.\nImportant: You must enter exactly " <> ToString[
     Length[votes]] <> " items.\nExample: {p, s, t, r, s}"], {i, 1, Length[cand]}];


isValid = True;
errorMsg = "";

If[Length[Transpose[preferance]] =!= Length[votes], isValid = False;
  errorMsg = "Error: Dimension Mismatch. The length of your
    preference rows does not match the number of voter groups."];

If[isValid, Do[If[Sort[Transpose[preferance][[v]]] =!= Sort[cand], isValid = False;
    errorMsg = "Error: Invalid Preference Table! Voter group " <> ToString[v] <>
      " does not contain a valid ranking of the exact candidates provided.";
    Break[];], {v, 1, Length[votes]}]];


If[isValid, first = preferance[[1]];
 Print[Style["Valid preference table built successfully!", Darker[Green], Bold]];
 Print["Your entered preferance table is:\n----------"];
 Print[TableForm[Prepend[preferance, votes]]], Print[Style[errorMsg, Red, Bold]];
 Abort[];]



(*Plurality Method*)

Print["\n\n\n\n\nPlurality Method\n-------------"];
firstvotes = Table[0, Length[cand]];
Do[firstvotes[[i]] = Total[votes[[Flatten[Position[first, cand[[i]]]]]]],
 {i, 1, Length[cand]}]
Print["Number of gained first count votes= ", firstvotes,
 "\n", Style[Row[{"Hence winner by Plurality Method is: ",
    cand[[Position[firstvotes, Max[firstvotes]][[1, 1]]]]}], Darker[Green], Bold]]
```

```
(*Bodra Count Method*)
Print["\n\n\n\n\nBorda count Method\n-------------"];
points = Table[0, Length[cand]];
Do[Do[points[[j]] +=
   Total[votes[[Flatten[Position[preferance[[i]], cand[[j]]]]]] * (Length[preferance] - i + 1)],
  {i, 1, Length[preferance]}], {j, 1, Length[cand]}]
Print["Number of Obtained points is= ", points,
 "\n", Style[Row[{"Hence winner by Borda-Count Method is: ",
    cand[[Flatten[Position[points, Max[points]]]]][[1]]}], Darker[Green], Bold]]





(*Plurality with Elimination Method*)

PluralityWithElimination[cands_, vts_, pref_] := Module[
   {activeCands = cands, voterProfiles = Transpose[pref], firstVotes, minVotes,
    elimCands, majority = Total[vts] / 2.0, winner = Null, round = 1, roundTable},

   Print["\n\n\n\n\nPlurality with Elimination Method\n-----------------"];
   While[Length[activeCands] > 0, firstVotes =
     Table[Total[vts[[Flatten[Position[voterProfiles[[All, 1]], c]]]]], {c, activeCands}];
    Print["Round ", round, " first-place vote counts:"];

    roundTable = Prepend[Table[{activeCands[[i]], firstVotes[[i]]},
       {i, 1, Length[activeCands]}], {"Candidate", "Votes"}];
    Print[Grid[roundTable, Dividers → {None, {False, True, False}}]];

    If[Max[firstVotes] > majority, Print["\nMajority achieved in Round ", round, "!"];
     winner = activeCands[[Position[firstVotes, Max[firstVotes]][[1, 1]]]];
     Break[];];


    minVotes = Min[firstVotes];
    elimCands = activeCands[[Flatten[Position[firstVotes, minVotes]]]];
    Print["\nEliminating (fewest votes = ", minVotes, "): ", elimCands, "\n"];
    activeCands = Complement[activeCands, elimCands];
    voterProfiles = Map[DeleteCases[#, Alternatives @@ elimCands] &, voterProfiles];
    round++;];
   Print["\n",
    Style[Row[{"Hence winner by Plurality with Elimination Method is: ", winner}],
     Darker[Green], Bold]];

  ];

PluralityWithElimination[cand, votes, preferance]
```

```
(*Pairwise Comparison Method*)

PairwiseComparison[cands_, vts_, pref_] := Module[
  {voterProfiles, numCands, numVoterGroups, wins, losses, c1,
   c2, rank1, rank2, p1Votes, p2Votes, maxWins, winners, summaryTable},
  voterProfiles = Transpose[pref];
  numCands = Length[cands];
  numVoterGroups = Length[vts];
  wins = Table[0, {numCands}];
  losses = Table[0, {numCands}];


  Print["\n\n\n\n\nPairwise Comparison Method\n-------------"];
  Do[

   c1 = cands[[i]];
   c2 = cands[[j]];
   p1Votes = 0;
   p2Votes = 0;

   Do[rank1 = Flatten[Position[voterProfiles[[v]], c1]][[1]];
    rank2 = Flatten[Position[voterProfiles[[v]], c2]][[1]];
    If[rank1 < rank2, p1Votes += vts[[v]], p2Votes += vts[[v]]], {v, 1, numVoterGroups}
   ];

   If[p1Votes > p2Votes, wins[[i]] += 1;
    losses[[j]] += 1, If[p2Votes > p1Votes, wins[[j]] += 1;
     losses[[i]] += 1, wins[[i]] += 1 / 2;
     wins[[j]] += 1 / 2 (*0.5 wins for a tie*)]], {i, 1, numCands - 1}, {j, i + 1, numCands}

  ];


  summaryTable = Prepend[Table[{cands[[i]], wins[[i]], losses[[i]]}, {i, 1, numCands}],
    {"Candidate", "Wins", "Losses"}];

  Print["Pairwise win/loss summary:"];
  Print[Grid[summaryTable, Dividers → {None, {False, True, False}}]];
  maxWins = Max[wins];
  winners = cands[[Flatten[Position[wins, maxWins]]]];
  Print["\n"];
  If[maxWins == numCands - 1 && Length[winners] == 1, Print["Condorcet winner found!"],
```

```
    Print["No Condorcet winner. Candidate with most pairwise wins:"]];

   Print[
    Style[Row[{"Hence winner by Pairwise Comparison Method is: ", Row[winners, ", "]}],
     Darker[Green], Bold]];];
```

```
PairwiseComparison[cand, votes, preferance]
ClearAll["Global`*"]
```

Valid preference table built successfully!

Your entered preferance table is:

1	2
a	b
b	a

Plurality Method

Number of gained first count votes= {1, 2}

Hence winner by Plurality Method is: b

Borda count Method

Number of Obtained points is= {4, 5}

Hence winner by Borda-Count Method is: b

Plurality with Elimination Method

Round 1 first-place vote counts:

Candidate	Votes
a	1
b	2

Majority achieved in Round 1!

Hence winner by Plurality with Elimination Method is: b

Pairwise Comparison Method

Pairwise win/loss summary:

Candidate	Wins	Losses
a	0	1
b	1	0

Condorcet winner found!

Hence winner by Pairwise Comparison Method is: b